Data Quality and Cleansing Report: Digital Usage

Project: University Library Analytics Data Warehouse

Data Source: `digital_usage.xlsx`

1. Executive Summary

The raw `digital_usage.xlsx` dataset contained several data quality issues that prevented accurate analysis. The ETL pipeline (built in Python/Pandas) was designed to extract, transform, and load this data into the Data Warehouse staging area. This report documents the specific issues found and the transformation rules applied to resolve them.

2. Identified Data Issues & Resolutions

Issue 1: Inconsistent Faculty Naming Conventions

Description: The Faculty column contained abbreviations and alternative spellings (e.g., 'ENG', 'Engr', 'Engineering', 'CS', 'CompSci').

Resolution: Created a mapping dictionary in Python to standardize all variations into clean, full names.

Example: 'ENG' and 'Engr' were mapped to 'Engineering'. 'CS' and 'CompSci' were mapped to 'Computer Science'.

Issue 2: Inconsistent Resource Type Capitalization

Description: The same resource type was logged differently across records (e.g., 'E-book', 'e-Book', 'E-BOOK', 'JOURNAL', 'journal').

Resolution: Applied string formatting rules. All variations of electronic books were standardized to 'E-book', and all variations of journals were standardized to 'Journal'.

Issue 3: Mixed Date Formats

Description: The Date column contained multiple formats, including standard ISO format (2024-01-15) and US format (01/15/2024).

Resolution: Used Pandas `pd.to_datetime` to parse the dates and then converted them to a strict ISO 8601 format (YYYY-MM-DD) for the Data Warehouse date dimension.

Issue 4: Missing Values in Duration_Minutes

Description: Approximately 23% of the records had NULL or empty values for Duration_Minutes (mostly older legacy records).

Resolution: Instead of imputing fake data or deleting the records (which would lose valuable download counts), we retained the NULL values and loaded them as 0 in the fact table, while keeping the original data flagged in staging.

Issue 5: Duplicate Records

Description: The system contained exact duplicates where the same user downloaded the same resource on the same day.

Resolution: Applied a `drop_duplicates` rule based on the combination of Date, UserType, ResourceType, and Faculty.

Result: Successfully removed 10 duplicate rows.

Issue 6: Download Count Data Types

Description: Some DownloadCount values contained commas (e.g., "1,200") which caused them to be treated as text instead of numbers.

Resolution: Stripped commas using string replacement and cast the column to an Integer (`int`) data type.

3. Before vs. After Metrics

Metric	Before ETL	After ETL
Total Rows	400	390
Duplicate Rows	10	0
Mixed Date Formats	Yes	Standardized (YYYY-MM-DD)
Faculty Codes	Unstandardized	Cleaned & Standardized

4. Conclusion

After the ETL process, the digital usage data is 100% accurate, deduplicated, and properly structured. It is now successfully linked to the `dim_date` and `dim_resource` dimensions in the Star Schema, ready for OLAP reporting.